Conclusion:

Question:

How has the rise in three-point shooting by NBA big men (PF-C) between 2000-2001 and 2025-2026 seasons related to team pace, offensive efficiency, and ball movement within the NBA?

League-Level(the big picture):

Change is always a factor no matter what kind of sport it is. Players get better, rules change, and overall style of play changes. In the early years of data big men didn't shoot many threes, they shot a very slim amount to about a quarter as time went on throughout the NBA. Their share of three pointers among their respected teams went from about 12% in 2001 to 26% in 2026, with a peak of 30% in the year 2020. Among qualified bigs (500+ minutes), the share averaging at least one three pointer a game rose from about 12% to about 72%. The power forward position (PF) has averaged at least one attempt per game every season since 2013 and even got to about 3.4 attempts a game by 2020. For a centers perspective they didn't stay above one three per game until 2019 and averaged about 1.9 in 2026. Over that same period, league pace rose from about 91 to about 99 possessions per 48 minutes. On top of that, offensive rating rose from about 103 to about 116, and team assists rose from around 22 to 27 per game. The biggest changes in all of these realms of data came between 2014 and. 2020. Since 2020, big-man share has been statistically flat while offensive rating kept rising, so the most recent offensive gains came from somewhere else as mentioned earlier.

Why these graphs: (How to read them)

- Trends: (bigs share of threes, pace, offensive rating, 2001-2026)

I used line charts, because time is on the horizontal axis and a line shows when things changed. All three of these rose, with the biggest jump between 2014 and 2020. This shows they changed at the same time, not that one caused another.

- Power forwards vs. Centers:

Two lines on one chart(PFs and Cs) so the gap is easy to see. Power Forwards started shooting more threes around 2013, and centers followed later on.

- Era scatter plots: (2001-2013 vs. 2014-2026)

Each dot is one team in one season, and the gray line is the average relationship. Both axes are measured against that season's league average, so the chart compares teams with their peers instead of just showing everything rising over time. The line tilts slightly upward in the early-era and is flat after 2014.

- Season slider:

The same scatter plot for one season at a time. This checks whether the pattern holds year by year. With only about 30 teams per season, any single year is noisy, so I have come to rely on the pool chart.

(I used lines for change over time and scatter plots for whether two measures move together across teams.)

Between Teams:

Because everything pretty much rose over time, I compared each team to the league average for its own season. From 2001 to 2013, teams whose bigs took 10 percentage points more of the team's threes had offensive ratings about 0.5 points per 100 possessions higher (95% range 0.2 to 0.8, p=0.003) That is a small effect and fragile to mess with: removing Dallas (Mavericks) lowered the correlation from 0.15 to 0.10, and controlling for each team's overall three-point attempts rate cut the effect roughly in half to +0.24, in which I can't distinguish from zero (p=0.14). From 2014 to 2026, I didn't find a relationship (-0.16, p=0.43). I found no efficient link between big-man three-point share and pace or assists in either period. From a basketball standpoint bigs shooting threes lead to longer easier to grab offensive rebounds, quicker shot taking, and as time goes on throughout the NBA skillsets become more reliable on three-pointers from every position including bigs. All of these different factors contribute to pace, efficiency, and assists that can and sometimes not show within statistical data.

Conclusion:

From the data I have collected there is not enough sufficient evidence that big men shooting threes, as opposed to teams shooting threes in general, drove changes in pace, efficiency, or ball movement. The league-wide rise happened alongside quicker, more efficient offense, but a team-by-team comparison cannot separate the big men contribution from the league as a whole adopting the three-pointer at once.

Limitations:

- Team-level data cannot show cause. A strong offense may give its bigs more freedom to shoot, not the reverse.
- "Big" means listed position (PF, Centers) which include some wings who can be listed as Power Forwards but misses hybrid players who play other positions at once.
- A dip in 2011-2012 makes an appearance in big-man share, in Power Forwards' attempts, and in bigs shooting threes. I cannot explain it further with this data I have/used.
- The 500-minute cutoff, the one-three-per-game threshold, and 2013/2014 split were choices of mine that I thought would make the data more presentable and understandable. The early-era result held for split years from 2011 to 2016.
- The team comparison cannot say whether the league-wide rise helped offenses throughout these years since it affected every team at once.
- "Flow" is measured only by assists per game in this data. Assists capture one kind of ball movement. I didn't have touches, passes, or time of possession , which only exist from 2013-2014 and which my original plan mentioned. The claim about ball movement is narrower than the question.

- Defense is absent in which the league's pace and scoring depend on how other teams defend throughout the league, which I didn't model.
- I ran several checks within the data after the first result and Dallas was chosen as the check because it stood out.
- Everything here is about teams , so it says nothing about individual players. A team's big-man share could be one or multiple players (PF and C).
- Pace and offensive ratings are estimates, leading to possessions aren't directly counted in box score data which I don't have so both rest on a published estimate.
- For traded players, the share uses each team stint separately, so a player listed at Center on one team and Power Forward on another is controlled by stint, while the player-level tables use a single combined row.
- Team assists are calculated by me. They're the sum of players' assists, not a published team figure in which it should all still match as if I did pull a team's published total.
- I picked 2013/2014 because the trend table showed a significant jump there. The result held for other years from 2011 to 2016 but this choice wasn't something that was made in advance.
- The same franchise appears about 26 times, and a team's offense one year tends to usually resemble its offense the next. This makes the p-values appear more convincing than they should since the average calculation assumes each row is unrelated. In this scenario I should have clustered the errors by team, which I didn't do so the early-era p=0.003 is more than likely too small but treated as less certain than it reads.
- .500 total minutes relates to different things in the shortened 2012 (66-game), 2020 (72-game, then the bubble), and 2021 (72-game) seasons.